LESSON PLAN

A/C ENGINEERING SYSTEM (17)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	23
Total No. of Lesson Plans	30	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Control system components
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Control system components
5. Specific Objectives	17.1.5.1 Trainee will be able to explain the construction and functioning of aircraft flight controls systems and its components.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of control system diagrams in IB; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three primary flight control surfaces?
(b) How does the pilot's input translate to control surface movement?
(c) Why is a control system necessary for safe flight?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Hydraulic power control systems • Mechanical linkage systems • Cable and push-pull rod systems • Control surface construction • Actuator types and functions	L & Ds	PPT, IB	15 min
(b)	• Flight control system components • Control horns and tabs • Bellcranks and pulleys • Hydraulic valves and pumps • Safety mechanisms and locks	L & Ds	PPT, PDF	20 min
(c)	• Functioning of control systems • Force transmission paths • System redundancy features • Maintenance access points • Common failure modes	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of hydraulic actuators in modern flight control systems.
(b) Describe how mechanical linkages ensure backup control authority.
(c) Identify two safety mechanisms used in flight control systems.

12. Home Assignment and Follow Up	Prepare a brief note on the need for simple oxygen systems and the effects of lack of oxygen, as covered in LP 24.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Need for simple oxygen system and pressurized cabin effect of lack of o2, Types of 02 regulators. Fire detection and protection system:	Refer to the master syllabus reference list	LP 24

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Engineer, P No. 12345	Senior Instructor, Chief Engineer, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min